Incident Response Lifecycle

Introduction

Incident Response Lifecycle is a structured process used by Security Operations Center (SOC) to handle security incidents such as malware infections, unauthorised access, data breaches, and ransomware attacks. The purpose of incident response is to detect incidents quickly, contain the damage, remove the threat, recover systems, and prevent future incidents.

The Incident Response Lifecycle is defined by the NIST SP 800-61 Incident Handling Guide and is widely used in cybersecurity operations.

Incident Response Lifecycle Phases

The Incident Response Lifecycle consists of six main phases:

Phase	Description
Preparation	Preparing tools, policies, and procedures
Identification	Detecting and confirming a security incident
Containment	Isolating affected systems to stop the attack
Eradication	Removing malware or attacker access
Recovery	Restoring systems to normal operation
Lessons Learned	Documenting incident and improving security

1. Preparation Phase

The preparation phase involves setting up security tools, creating incident response policies, and training the security team. This phase ensures the organisation is ready to respond to security incidents.

Examples:

- Install security tools (SIEM, IDS, EDR)
- Create incident response playbooks
- Backup critical data
- Security awareness training
- Define communication plan

2. Identification Phase

In this phase, security analysts monitor alerts and logs to identify whether a security incident has occurred. Analysts investigate alerts, collect logs, and determine if the alert is a false positive or a real incident.

Examples:

- SIEM alert triggered
- Suspicious login detected
- Malware detected by antivirus
- Unusual network traffic detected

3. Containment Phase

The containment phase focuses on stopping the attack from spreading to other systems. The affected systems are isolated from the network.

Examples:

- Disconnect infected system from network
- Block malicious IP address
- Disable compromised user account
- Stop malicious processes

Containment can be:

- Short-term containment
- Long-term containment

4. Eradication Phase

In this phase, the root cause of the incident is removed from the system.

Examples:

- Remove malware
- Delete malicious files
- Patch vulnerabilities
- Remove attacker backdoor access
- Reset compromised passwords

5. Recovery Phase

In the recovery phase, systems are restored to normal operation and monitored to ensure the attack does not happen again.

Examples:

- Restore files from backup
- Reconnect system to network
- Monitor system logs
- Test system functionality

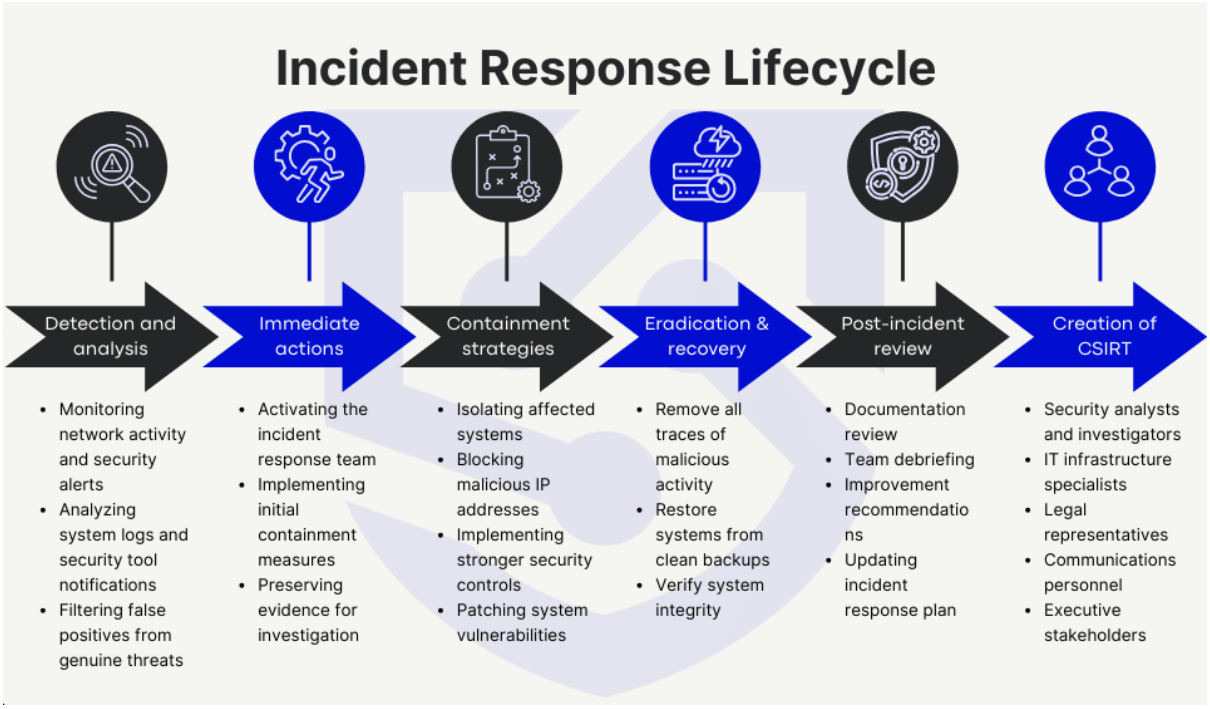
6. Lessons Learned Phase

This is the final phase where the incident is documented and improvements are made to prevent similar incidents in the future.

Examples:

- Write incident report
- Update security policies
- Improve monitoring rules
- Conduct team meeting
- Update incident response plan

Incident Response Lifecycle Diagram



Example Incident Response Flow

Example: Malware Infection Incident

Phase	Action
Identification	Malware alert detected
Containment	Infected system isolated
Eradication	Malware removed
Recovery	System restored
Lessons Learned	Security policy updated

Conclusion

The Incident Response Lifecycle provides a structured approach to handling security incidents. It helps organisations detect incidents quickly, minimise damage, recover systems, and improve security processes for the future.